Impact of Feature Selection and Genetic Algorithms on Machine Learning Performance for Intraday Financial Trend Prediction

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Abstract: This study examines the effect of feature selection and Genetic Algorithm-based hyperparameter optimization on intraday financial trend forecasting in the Brazilian futures market. Using Mini Index (WIN) contracts from the BovDBv2 database, five feature selection methods were evaluated: Information Gain, ReliefF, PCA, ClassifierAttributeEval, and CorrelationAttributeEval. Random Forest and Multilayer Perceptron classifiers were employed, with optimal feature subsets selected through stratified cross-validation and model hyperparameters subsequently optimized by a Genetic Algorithm. All optimized Random Forest models outperformed the baseline accuracy of 0.6381, with PCA achieving the best RF result at 0.6555. The best overall performance was obtained by the MLP with PCA and GA-tuned hyperparameters, reaching 0.6571. These findings indicate that jointly improving feature representation and model configuration enhances generalization in noisy and nonstationary market prediction tasks.

1 INTRODUCTION

Intraday financial forecasting aims to identify short-term patterns in high-frequency market data, a task characterized by noise, nonstationarity, and weak predictive signals. In the Brazilian market, Mini Index futures (WIN) are widely traded due to their high liquidity, making them a relevant benchmark for intraday prediction models.

Machine Learning (ML) techniques have been increasingly applied to financial forecasting, as they can capture complex nonlinear relationships from historical data Usmani et al. (2016). In this context, technical indicators derived from price and volume series are commonly used as input features. However, the large number of correlated indicators may introduce redundancy and degrade generalization performance.

Feature selection methods aim to mitigate this issue by identifying the most informative variables and reducing dimensionality Kumar and Minz (2014). In parallel, hyperparameter optimization plays a critical role in improving model performance, particularly in flexible models such as Random Forest (RF) and Multilayer Perceptron (MLP). Evolutionary approaches, such as Genetic Algorithms (GA), are well suited for this task due to their ability to explore complex and non-convex search spaces Goldberg (1989).

This work addresses this gap by conducting a controlled empirical study on the joint effect of feature selection and GA-based hyperparameter optimization for intraday trend prediction. The main contributions are threefold: (i) the integration of feature selection and evolutionary optimization within a unified experimental framework, (ii) an evaluation using high-frequency Mini Index futures data from the Brazilian market, and (iii) an analysis of generalization behavior under different feature representations. Although the observed improvements in predictive performance are modest, they are consistent across configurations, which is particularly relevant in financial forecasting tasks.

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The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the proposed methodology. Section 4 presents the experimental results, and Section 5 concludes the paper.

2 RELATED WORK

Machine learning has become a central tool in financial market forecasting, with studies consistently reporting that models trained on technical indicator features outperform those relying on raw price series alone Nabipour et al. (2020).

MLPs are widely used due to their ability to capture nonlinear relationships without strong assumptions. Ecer et al. (2020) showed that architectural choices, such as network depth and activation functions, directly affect generalization performance. Recurrent models such as LSTMs have also been explored Kamalov et al. (2020), but their computational cost and sensitivity to hyperparameters limit their applicability in high-frequency settings. In contrast, RF has emerged as a robust baseline for financial classification, as it effectively handles noisy and heterogeneous features Ballings et al. (2015). However, its performance remains strongly dependent on feature representation and model configuration.

Technical indicators are commonly used as input features, but large set of indicators often introduce redundancy and multicollinearity, which can degrade model generalization Souza et al. (2025). Feature selection methods address this issue by identifying the most relevant attributes. For example, Naik and Mohan (2019) reported a 12% reduction in prediction error after applying feature selection to technical indicators, while Tanrikulu and Pabuccu (2024) showed that compact feature sets can outperform full indicator sets in cryptocurrency forecasting.

GA have been widely adopted for hyperparameter optimization due to their ability to explore complex and non-convex search spaces. Ding et al. (2011) demonstrated that hybrid GA-based strategies improve convergence in neural networks, while Guo (2025) showed that evolutionary tuning improves RF performance in large-scale financial datasets.

Despite these advances, feature selection and hyperparameter optimization are typically addressed as independent stages. This separation limits the understanding of how feature representation and model configuration interact, particularly in intraday forecasting scenarios. This work addresses this gap by evaluating both processes jointly within a unified experimental framework.

3 METHODOLOGY

This section presents the methodology adopted in this study. Figure 1 illustrates the proposed methodological process, highlighting the main stages of the approach developed in this work.

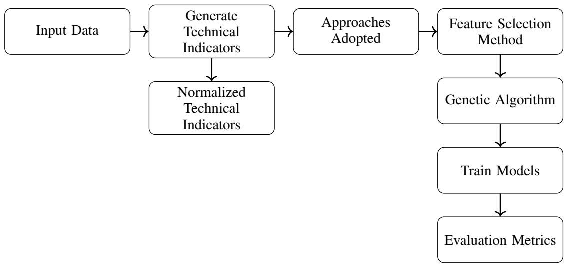


Figure 1: Overview of the proposed methodology.

3.1 Input Data

The data employed in this study were obtained from the BovDbV2[1] database Souza et al. (2024), an updated version of BovDb Cardoso et al. (2022) that provides publicly available, pre-processed data from (B3[2]) for research purposes. In addition to historical daily data, BovDbV2 includes intraday records sampled at 5-minute intervals from January to June 2024, enabling a more detailed examination of short-term market behavior.

This work focuses on futures contracts extracted from the intraday *Price5* table, with emphasis on the Mini Index futures contracts (WIN), a derivative linked to the Ibovespa index. These contracts were selected for their high liquidity and intense trading activity, which makes them particularly suitable for intraday forecasting. From these intraday series, technical analysis indicators were computed and used as explanatory variables in the predictive models.

3.2 Technical Indicators Construction

Technical indicators were computed from intraday price series and normalized to ensure scale compatibility. The feature set includes SMA and EMA, used to capture trend behavior Billah et al. (2024), as well as rolling standard deviation (STD), employed as a proxy for volatility Altman and Bland (2005). Momentum is modeled through MACD, defined from the difference between exponential moving averages Aguirre et al. (2020), while trend strength is represented by ADX and its smoothed variant ADXR Tharavanij et al. (2015).

[1] https://doi.org/10.7910/DVN/TMB4IG

[2] https://www.b3.com.br/

Short windows $w \in \{3, 5, 7, 9\}$ were adopted for SMA, EMA, and STD, while MACD parameters were adjusted to $(6, 13, 4)$ for intraday sensitivity. ADXR was computed as a smoothed version of ADX.

Normalization was applied according to feature characteristics:

- **Price-based features (Open, Close, SMA, EMA, STD):**
$$x_{\text{norm}} = \frac{x_t - \text{Low}_t}{\text{High}_t - \text{Low}_t} \quad (1)$$
- **ADX and ADXR:**
$$x_{\text{norm}} = \frac{x_t}{100} \quad (2)$$
- **MACD components:** rolling z-score (window = 100) followed by sigmoid scaling.

3.3 Genetic Algorithm Optimization

A Genetic Algorithm (GA) was used to optimize model hyperparameters after feature selection Goldberg (1989). The optimization followed a standard evolutionary process with tournament selection, crossover, mutation, and elitism.

The adopted configuration is summarized in Table 1.

Table 1: Genetic Algorithm configuration.

Parameter	Value
Generations (N_{gen})	1000
Population (N_{pop})	10
Crossover (p_c)	0.80
Mutation (p_m)	0.50
Mutation (bit)	0.05
Tournament size	3
Elitism	1
Seed	42

Fitness was computed on the training set using stratified cross-validation, while the test set was reserved for final evaluation. To balance fitting and generalization, the fitness function combines training and validation accuracy:

$$\text{Fitness} = 0.4 \cdot Acc_{train} + 0.6 \cdot Acc_{val} \quad (3)$$

The final fitness corresponds to the average across folds. After optimization, the best configuration was retrained on the full training set and evaluated on the test set.

3.4 Feature Selection and Model Training

Feature selection methods were applied to rank the input variables, including Information Gain Omuya et al. (2021), ReliefF Yu and Liu (2003), Principal Component Analysis (PCA) Omuya et al. (2021), ClassifierAttributeEval, and CorrelationAttributeEval. These methods were used to identify the most informative features while reducing redundancy and multicollinearity.

Experiments were conducted on intraday data from BovDBV2, sampled at 5-minute intervals. The input space consisted of normalized technical indicators derived from price series, as described in Section 3.2. For each selection method, features were incrementally added according to their ranking, allowing the evaluation of model performance as a function of subset size.

Two classifiers were considered: Random Forest (RF) and Multilayer Perceptron (MLP). RF was adopted due to its robustness to noise and ability to handle nonlinear relationships, providing a strong baseline. Model performance was assessed using stratified k-fold cross-validation on the training set, preserving class distribution across folds. In each iteration, the model was trained on $k-1$ folds and validated on the remaining fold, with the final score obtained as the average across all folds. The best configuration for each feature subset was then retrained on the full training set and evaluated on the test partition.

4 RESULTS

This section presents the main results of the study and discusses their implications. It is organized into three parts. The first subsection, 4.1, describes the experimental setup, including data preparation, organization, and partitioning from BovDBV2. The second subsection, 4.2, examines predictive results for the complete set of features and for subsets generated by feature selection methods, emphasizing the most informative attributes. Subsection 4.3 evaluates the models after Genetic Algorithm–based hyperparameter optimization and compares them under the same experimental protocol.

4.1 Experimental Setup and Data Partitioning

The prediction task considered in this study is formulated as a binary classification problem, where the target variable indicates the short-term market trend. A value of 0 denotes a downtrend, whereas a value of 1 denotes an uptrend. These labels were generated through a peak-and-trough identification procedure adopted from previous work Souza et al. (2025). To ensure a controlled training process, the training partition was balanced, containing 1,500 instances per class (3,000 samples in total). In contrast, the test partition preserved the original class distribution, with 5,712 downtrend instances and 6,486 uptrend instances, allowing the final evaluation to better reflect realistic market conditions.

The models were optimized and evaluated on the training partition using stratified 5-fold cross-validation. In each iteration, four folds were used for training and one for validation, preserving class proportions in every split. The validation accuracy corresponds to the average across all folds. After optimization, the best configuration for each model was retrained on the full training set and evaluated on the unseen test partition.

All experiments were conducted under the same computational environment to ensure a fair comparison of runtime and predictive performance. The experiments were carried out on a CPU-based system with 12 processing cores and 16 GB of RAM. Considering the feature selection methods evaluated in this study, the total execution times were 4h 00m 50s for ClassifierAttributeEval, 4h 51m 25s for CorrelationAttributeEval, 12h 33m 44s for Information Gain, 6h 21m 35s for PCA, and 4h 42m 37s for ReliefF. In addition, optimizing and training the MLP model required approximately 20 hours, confirming the higher computational cost of neural network optimization in this setting.

4.2 Performance Evaluation: Full Feature Set vs. Feature Selection

This section compares baseline models trained on the full feature set with configurations obtained through feature selection methods, aiming to assess the impact of reducing the input space on predictive performance and generalization.

4.2.1 Baseline: Random Forest

The Random Forest baseline was trained using all 31 features, including technical indicators and intraday variables described in Section 3.2. The model was configured with 100 trees and default parameters.

Using stratified 5-fold cross-validation, the model achieved an average accuracy of 0.6250, indicating stable performance during training. When evaluated on the test set, the model reached an accuracy of 0.6381 and an AUC-ROC of 0.6897, suggesting moderate discriminative capability. Overall, the results indicate balanced classification performance without significant overfitting, establishing a strong baseline for subsequent comparisons.

4.2.2 Baseline: Multilayer Perceptron

The baseline MLP consisted of a single hidden layer with 64 neurons, tanh activation, dropout (0.2), and L2 regularization. Training was performed using RMSprop with binary cross-entropy loss.

The model achieved an average cross-validation accuracy of 0.6213. On the test set, performance decreased slightly to 0.6085, with an AUC-ROC of 0.6406. Compared to the Random Forest, the MLP showed slightly lower predictive performance, although still maintaining stable generalization. These results serve as a neural baseline for evaluating the impact of feature selection and GA-based optimization.

4.2.3 Feature Selection Methods: Training Performance

In this step, each feature selection method described in Section 3.4 is applied to the training data to produce a ranked list of features. The Random Forest model is then retrained incrementally, adding one feature at a time According to each method's ranking, using the same hyperparameters from the baseline experiment. This procedure allows us to identify the point of maximum accuracy (k^*), defined as the feature subset size that achieves the highest cross-validation accuracy for each method.

Figure 2a shows the evolution of cross-validation accuracy as features are incrementally added for each method. All Methods follow a similar pattern: accuracy rises steeply in the first few features, stabilizes around $k = 9$–13, and plateaus or slightly declines beyond $k = 20$. Information Gain and ClassifierAttributeEval reaches its peak earlier and at a higher values, while PCA consistently lags behind the other methods across the entire range, suggesting that variance-based projection is less More effective than filter-based ranking for this task. The shaded bands represent ± 1 standard deviation across folds, indicating that All methods exhibit moderate variance, with overlapping confidence intervals regions in the plateau zone.

Table 2 presents the features selected by each method at k^*. Features appearing in three or more methods are highlighted in bold, indicating consensus across different selection criteria.

Figure 2b compares cross-validation and test accuracy at k^* for all methods. ClassifierAttributeEval showed the highest CV accuracy (0.6450, $k^* = 25$), while CorrelationAttributeEval led in test accuracy (0.6479, $k^* = 28$), outperforming the full feature set baseline (0.6381). ReliefF also exceeded the baseline on the test set (0.6415). The CV-test accuracy

Table 2: Summary of feature selection results at optimal subset size k^* for each method. Features appearing in three or more methods are considered consensus features.

Method	Consensus Features
Information Gain	15 of 23
ReliefF	18 of 28
PCA	16 of 29
ClassifierAttributeEval	16 of 25
CorrelationAttributeEval	17 of 28

Consensus features (present in ≥ 3 methods): `MACD_signal_norm`, `MACD_line_norm`, `MACD_hist_norm`, `ADXR_14_norm`, `ADX_14_norm`, `open_norm`, `close_norm`, `std_close9_norm`, `EMA_3_norm`, `EMA_7_norm`, `EMA_9_norm`, `SMA_3_norm`, `SMA_5_norm`, `SMA_9_norm`, `open`, `close`, `high`, `low`, `average`.

gap was consistently small (under 0.003), indicating stable generalization. PCA performed worst at both stages (0.6073 CV, 0.6065 test), confirming linear dimensionality reduction is less effective than filter-based feature ranking for this classification task.

4.3 Genetic Algorithm Hyperparameter Optimization and Benchmark

Following the feature selection experiments, the Genetic Algorithm described in Section 3.3 was applied to optimize the hyperparameters of the Random Forest classifier for each feature selection method, using the corresponding optimal subset of size k^*. Table 3 summarizes the hyperparameters and their respective search ranges. Each parameter is encoded as a binary string, whose length determines the search resolution, resulting in a total chromosome length of 29 bits. The remaining GA settings follow Table 1.

Table 3: Random Forest hyperparameters optimized by the Genetic Algorithm and their respective search spaces.

Hyperparameter	Range / Values	Bits
`n_estimators`	[50, 500]	9
`max_depth`	[5, 50]	6
`min_samples_split`	[2, 20]	5
`min_samples_leaf`	[1, 20]	5
`max_features`	$\{0.1, 0.2, \ldots, 1.0\}$	4

Table 4 reports the evaluation metrics obtained on the test set after GA-based hyperparameter optimization, compared to the respective baseline models. All RF configurations optimized by the GA outperformed the baseline Random Forest (Test Acc. = 0.6381), confirming that the evolutionary search identified more suitable hyperparameter regions. However, the absolute gains over the baseline are modest, with test accuracies ranging from 0.6432 (Information Gain) to 0.6555 (PCA).

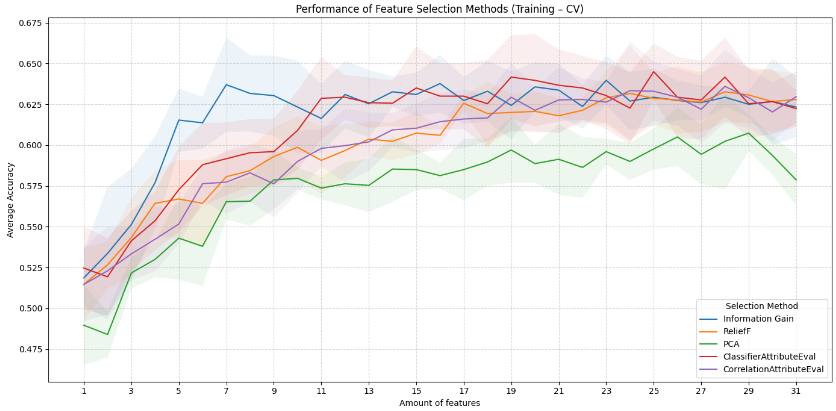
A consistent gap between cross-validation accuracy (≈ 0.79) and test accuracy (≈ 0.65) is observed across all RF variants. This behavior suggests that the GA, by optimizing fitness on training folds, tends to favor configurations that partially capture noise in the data. In particular, deeper trees increase model variance, which negatively impacts generalization. As a result, part of the performance gain obtained during optimization does not transfer to unseen data. This limitation is consistent with the known sensitivity of tree-based ensembles to high-variance configurations in noisy financial environments.

Among all feature selection strategies evaluated with the Random Forest, PCA achieved the best test accuracy (0.6555), as shown in Table 4. Furthermore, the fitness evolution curves in Figure 3a indicate stable convergence of the GA when PCA components are used as input features, with the best individual (HOF) consistently separating from the population mean across generations. This behavior suggests that the decorrelated and orthogonal representation produced by PCA provides a smoother search landscape for the evolutionary process. Based on these results, PCA was selected for the MLP experiments.

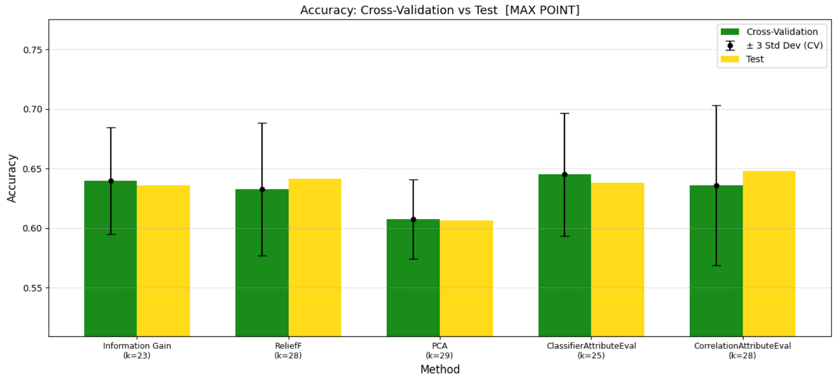
The GA was subsequently applied to optimize the MLP hyperparameters using the same $k^* = 29$ PCA components. The search space included architectural and regularization parameters, namely the number of hidden neurons, learning rate, L2 regularization, dropout rate, and batch size, encoded into a 30-bit chromosome. The same GA configuration was adopted, with early stopping patience set to 5 generations. Fitness was evaluated via stratified 5-fold cross-validation, training each candidate for 10 epochs per fold.

For the Random Forest, the GA consistently converged to configurations with low `max_features` (0.1) and minimal leaf size, indicating a preference for high-variance trees compensated by ensemble averaging. For the MLP, the best configuration included 87 hidden neurons with `tanh` activation, a learning rate of 4.64×10^{-3}, L2 regularization of 1.45×10^{-7}, and a dropout rate of 0.02, reflecting a balance between model capacity and regularization.

The results in Table 4 show that the proposed pipeline produced consistent improvements over the baseline models, although the magnitude and nature of these gains differed across classifiers. For the Random Forest, all GA-optimized variants outperformed the baseline test accuracy of 0.6381, with gains of up to 1.74 percentage points. However, these improvements were accompanied by a substantial gap between cross-validation and test performance, indi-



(a) Cross-validation accuracy as a function of the number of selected features for each method. Shaded regions represent ± 1 standard deviation across folds.



(b) Cross-validation and test accuracy at the point of maximum CV accuracy (k^*) for each feature selection method. Error bars represent ± 3 standard deviations across folds.

Figure 2: Summary of results obtained with the evaluated feature selection methods.

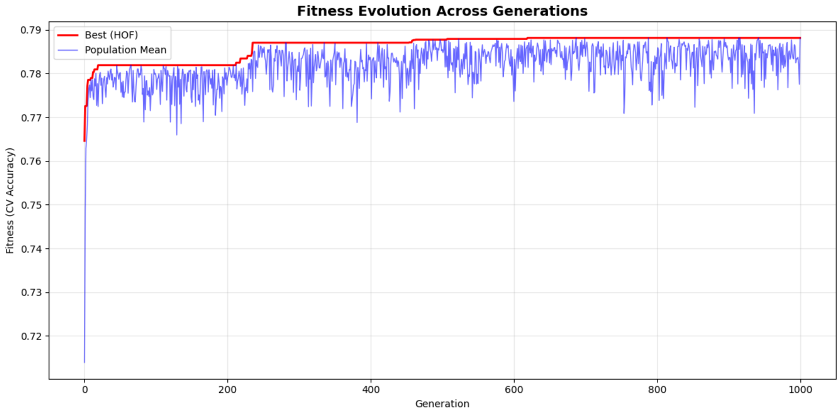
Table 4: Evaluation metrics on the test set after Genetic Algorithm hyperparameter optimization, compared to baseline models.

Method	k^*	**CV Acc.**	**Test Acc.**	**Precision**	**Recall**	**F1**
Baseline RF (no FS, no GA)	31	0.6250	0.6381	0.64	0.64	0.64
Information Gain + GA (RF)	23	0.7934	0.6432	0.64	0.64	0.64
ReliefF + GA (RF)	28	0.7922	0.6546	0.65	0.65	0.65
PCA + GA (RF)	**29**	**0.7881**	**0.6555**	**0.66**	**0.66**	**0.66**
ClassifierAttributeEval + GA (RF)	25	0.7917	0.6508	0.65	0.65	0.65
CorrelationAttributeEval + GA (RF)	28	0.7911	0.6505	0.65	0.65	0.65
Baseline MLP (no FS, no GA)	31	0.6213	0.6085	0.61	0.61	0.61
PCA + GA (MLP)	**29**	**0.6474**	**0.6571**	**0.66**	**0.66**	**0.66**

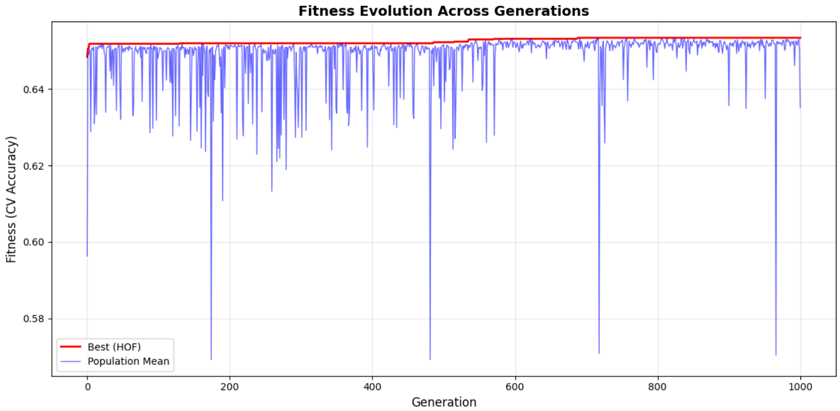
cating that part of the fitness increase obtained during the evolutionary search did not fully transfer to unseen data. This suggests that, for tree-based ensembles, hyperparameter optimization alone is not sufficient to eliminate the tendency to adapt too closely to noisy patterns in financial time series, even after feature selection.

A more substantial improvement is observed for the neural approach. The MLP trained on PCA-transformed features and optimized by the GA achieved the best overall test accuracy (0.6571), outperforming the baseline MLP (0.6085) by 4.86 percentage points and slightly surpassing the best Random Forest configuration.

Although the margin over RF is relatively small, this result is methodologically relevant, as it indicates that performance gains are not solely due to hyperparameter tuning. Instead, they emerge from the inter-



(a) GA fitness evolution for Random Forest with PCA.



(b) GA fitness evolution for MLP with PCA.

Figure 3: Fitness evolution of the Genetic Algorithm. The red curve represents the best individual (HOF), while the blue curve indicates the population fitness.

Table 5: Best MLP+PCA hyperparameters found by the Genetic Algorithm.

Hyperparameter	**Value**
Neurons (layer 1)	87
Learning rate	4.640×10^{-3}
L2 regularization (α)	1.451×10^{-7}
Dropout rate	0.02
Batch size	48
Activation function	`tanh`
Optimizer	RMSprop

action between a compact, decorrelated feature space and a model whose capacity is sensitive to optimization. In this sense, PCA not only reduces dimensionality but also reshapes the input space in a way that improves stability and generalization.

Taken together, these findings indicate that predictive performance in financial forecasting depends not only on the choice of model, but also on the interaction between feature representation and hyperparameter configuration. The superior performance of the MLP+PCA+GA combination suggests that evolutionary optimization is particularly effective for models sensitive to feature scale and correlation. Rather than identifying a single best model, the main contribution of this benchmark is to show that jointly addressing feature preprocessing and hyperparameter optimization leads to more robust generalization in noisy, non-stationary market environments.

5 CONCLUSIONS

This study evaluated the joint impact of feature selection and Genetic Algorithm (GA)–based hyperparameter optimization on intraday financial trend prediction. The results show that combining dimensionality reduction with evolutionary tuning consistently improves predictive performance compared to baseline configurations, although the magnitude of these gains remains inherently limited by the noisy and weakly predictable nature of short-term financial data.

Among the evaluated approaches, PCA-based representations combined with GA optimization achieved the best overall results, indicating that reducing feature redundancy and decorrelating input variables plays a critical role in enhancing model generalization. Although the observed gains over baseline models are modest in absolute terms, they are consistent across experimental settings and align with the well-known difficulty of achieving significant improvements in intraday market prediction tasks. In this sense, the main contribution of this work lies in demonstrating that a unified optimization pipeline can systematically extract incremental predictive signal under realistic market conditions.

Future work includes extending this framework to more advanced architectures, such as recurrent and attention-based models, as well as incorporating transaction costs and rolling evaluation protocols to better assess real-world applicability.

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REFERENCES

Aguirre, A. A. A., Medina, R. A. R., and Méndez, N. D. D. (2020). Machine learning applied in the stock market through the moving average convergence divergence (macd) indicator. *Investment Management & Financial Innovations*, 17(4):44–55.

Altman, D. G. and Bland, J. M. (2005). Standard deviations and standard errors. *BMJ*, 331(7521):903.

Ballings, M., Van den Poel, D., Hespeels, N., and Gryp, R. (2015). Evaluating multiple classifiers for stock price direction prediction. *Expert Systems with Applications*, 42(20):7046–7056.

Billah, M. M., Sultana, A., Bhuiyan, F., and Kaosar, M. G. (2024). Stock price prediction: Comparison of different moving average techniques using deep learning model. *Neural Computing and Applications*, 36(11):5861–5871.

Cardoso, F. C., Malska, J. A. V., Ramiro, P. J., Lucca, G., Borges, E. N., de Mattos, V. L. D., and Berri, R. A. (2022). Bovdb: A data set of stock prices of all companies in b3 from 1995 to 2020. *Journal of Information and Data Management*, 13(1).

Ding, S., Su, C., and Yu, J. (2011). An optimizing bp neural network algorithm based on genetic algorithm. *Artificial Intelligence Review*, 36(2):153–162.

Ecer, F., Ardabili, S., Band, S. S., and Mosavi, A. (2020). Training multilayer perceptron with genetic algorithms and particle swarm optimization for modeling stock price index prediction. *Entropy*, 22(11):1239.

Goldberg, D. E. (1989). *Genetic Algorithms in Search, Optimization and Machine Learning*. Addison-Wesley, Reading, MA.

Guo, P. (2025). Empirical study of the effectiveness and optimization of technical indicators in stock markets. Master's thesis, University of California, Los Angeles.

Kamalov, F., Smail, L., and Gurrib, I. (2020). Forecasting with deep learning: S&p 500 index. In *Proceedings of the 13th International Symposium on Computational Intelligence and Design (ISCID)*, pages 422–425. IEEE.

Kumar, V. and Minz, S. (2014). Feature selection: A literature review. *Smart Computing Review*, 4(3):211–229.

Nabipour, M., Nayyeri, P., Jabani, H., and Mosavi, A. (2020). Predicting stock market trends using machine learning and deep learning algorithms via continuous and binary data: A comparative analysis. *IEEE Access*, 8:150199–150212.

Naik, N. and Mohan, B. R. (2019). Optimal feature selection of technical indicators and stock prediction using machine learning techniques. In *Proceedings of the International Conference on Emerging Technologies in Computer Engineering*, pages 261–268. Springer.

Omuya, E. O., Okeyo, G. O., and Kimwele, M. W. (2021). Feature selection for classification using principal component analysis and information gain. *Expert Systems with Applications*, 174:114765.

Souza, A. S., Lucca, G., Borges, E. N., Cardoso, F. C., Dalmazo, B. L., and Berri, R. (2024). Dataset for intraday analysis of b3 stock prices. Harvard Dataverse.

Souza, A. V. S., Pinto, R. F., Dalmazo, B. L., Borges, E. N., Lucca, G., Mattos, V. L. D., and Berri, R. A. (2025). Predictive analysis with technical indicators and feature selection for futures contracts trading. In *Proceedings of the International Conference on Computational Science and Its Applications*, pages 349–367. Springer.

Tanrikulu, H. M. and Pabuccu, H. (2024). The effect of data types on the performance of machine learning algorithms for financial prediction. *arXiv preprint arXiv:2404.19324*.

Tharavanij, P., Siraprapasiri, V., and Rajchamaha, K. (2015). Performance of technical trading rules: Evidence from southeast asian stock markets. *SpringerPlus*, 4(1):552.

Usmani, M., Adil, S. H., Raza, K., and Ali, S. S. A. (2016). Stock market prediction using machine learning techniques. In *Proceedings of the 3rd International Conference on Computer and Information Sciences (ICCOINS)*, pages 322–327. IEEE.

Yu, L. and Liu, H. (2003). Feature selection for high-dimensional data: A fast correlation-based filter solution. In *Proceedings of the 20th International Conference on Machine Learning (ICML)*, pages 856–863.